

that the radiation is the intended one. To achieve this, the laser target designator and the laser seeker used in the guided-weapon delivery mission use the same PRF code, and the PRF code compatibility check forms the basis of identification of the desired radiation. PRF code compatibility is therefore essential to the weapon's functionality and mission success. The code is generally chosen to an accuracy of $\pm 1\mu\text{s}$ to $\pm 2\mu\text{s}$ in the time interval between two successive laser pulses in a nominal value that is usually in the range of 50ms to 200ms.

The opto-electronic position sensor in the front-end of the seeker unit determines the orientation of the weapon with respect to the target. But before it does that, it deciphers the PRF of the received radiation, and only if the PRF matches with the chosen PRF value within the specified tolerance, it is further processed to extract information on the angular position of the weapon with respect to the target. Information on angular error in azimuth and elevation is fed to the servo control sub-system in the specified format, which, in turn,

controls the flight trajectory of the weapon with the help of front canards and a tail unit.

The position sensor

The opto-electronic sensor employed for the purpose of position sensing is usually a quadrant photo sensor. It is the heart of the laser seeker. A two-dimensional array of photo sensors is also used in some cases. Fig. 3 explains the principle of operation of the quadrant photo sensor when used for the position-sensing application in laser-guided munitions.

The quadrant photo sensor is placed before the focal plane of front-end optics. The focal spot is symmetrical to the centre of the quadrant photo sensor when the perpendicular to focal plane of the detector is collinear with the axis of received laser radiation scattered from the intended target as shown in Fig. 4(a). This is the case when the weapon is pointing precisely towards the target.

If the laser radiation is impinging on the laser seeker cross-section at an angle, which will be the case when the weapon is not pointing towards the intended target, the centre of the focused laser spot will shift depending upon angular error in azimuth and elevation as shown in Figs 4(b) to 4(e).

The beam position in X

and Y directions is calculated using the following equations; X and Y represent angular errors in azimuth and elevation directions:

$$X = \frac{(B+C) - (A+D)}{(A+B+C+D)} \quad \text{and} \quad Y = \frac{(A+B) - (C+D)}{(A+B+C+D)}$$

where A, B, C and D are electrical voltages corresponding to laser power falling on the four quadrants. In the case of precise pointing towards the target, all four power levels are equal, and therefore $X=0$ and $Y=0$. $A+B+C+D$ represents total power, and division by total power ensures that the calculated position error is independent of laser-intensity variations.

Output analogue signals proportional to the magnitude of laser power falling on four quadrants are digitised and then processed to compute X and Y. Error signals X and Y are then used to guide the weapon towards the desired position of null with canards driven by a servo control system. The maximum value of the proportional field-of-view offered by the laser seeker of this type in X and Y directions is proportional to $\pm R$ where R is the radius of the focused laser spot. Larger spot size gives a larger field-of-view but a lower angular resolution. Radius of the focused spot can at the most be equal to half of the radius of the quadrant active area.

In another position sensing technique that uses a two-dimensional array based sensor, each active element in the array is identified by a unique azimuth and elevation angle. The focused spot at any time covers more than one active element, and the angular error in azimuth and elevation is computed from following equations. Fig. 5 explains the concept.

$$X = n \times \theta + \frac{[(B+C) - (A+D)]}{[(A+B+C+D)]} \quad \text{and}$$

$$Y = m \times \theta + \frac{[(A+B) - (C+D)]}{[(A+B+C+D)]}$$

where θ is the proportional field-of-view of each miniature quadrant. n and m are constants, depending

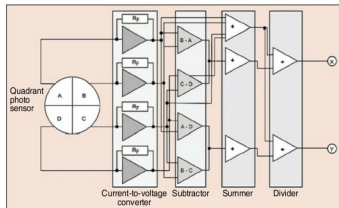


Fig. 3: Principle of operation of a quadrant photo sensor

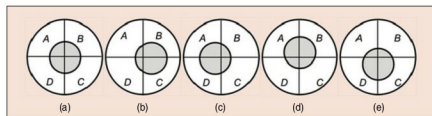


Fig. 4: Position of a laser beam spot with (a) laser beam aligned with the receiver's optical axis, (b) laser beam shifted in positive X-direction, (c) laser beam shifted in negative X-direction, (d) laser beam shifted in positive Y-direction and (e) laser beam shifted in negative Y-direction